Network Slicing in 5G New Radio: Architecture, Enabling Technologies, and Use Cases

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***Abstract*—Network slicing is a foundational technology in 5G New Radio (NR) that enables the creation of multiple logically independent, end-to-end virtual networks over a shared physical infrastructure. Each slice is tailored to meet the strict quality-of-service requirements of a specific application or vertical industry. This paper presents a comprehensive overview of network slicing in 5G NR, covering its core architecture, the enabling technologies that make it feasible, and the principal use cases driving its adoption. We examine how Software-Defined Networking, Network Function Virtualization, and Multi-Access Edge Computing combine to support dynamic slice orchestration. We also discuss the key challenges of inter-slice isolation, resource management, and security that must be addressed for wide-scale deployment.**

***Index Terms*—5G NR, network slicing, software-defined networking, network function virtualization, multi-access edge computing, resource management, eMBB, URLLC, mMTC**

I. Introduction

The fifth generation of mobile networks (5G New Radio) represents a paradigm shift from its predecessors by moving beyond a one-size-fits-all connectivity model toward a service-centric architecture. Unlike 4G Long Term Evolution (LTE), which was engineered primarily to deliver broadband data, 5G is designed to simultaneously serve vastly different application domains—from ultra-high-definition video streaming requiring multi-gigabit throughput, to factory automation demanding sub-millisecond latency, to smart-city sensor grids connecting millions of low-power devices per square kilometer [1].

Meeting these heterogeneous requirements over a single physical infrastructure would be impossible without a mechanism for logical partitioning. Network slicing provides exactly that mechanism. Standardized by the 3rd Generation Partnership Project (3GPP) in Release 15 and refined through Releases 16 and 17, network slicing allows operators to carve out isolated, end-to-end virtual networks—called *slices*—each with dedicated control-plane and user-plane functions, its own quality-of-service (QoS) policies, and independent lifecycle management [2].

The importance of network slicing extends beyond technical elegance. It is also a business enabler, allowing operators to offer differentiated service-level agreements (SLAs) to vertical industries such as healthcare, automotive, manufacturing, and entertainment, effectively transforming the network into a platform for innovation [3]. This paper is organized as follows: Section II describes the 5G network slicing architecture. Section III discusses the enabling technologies. Section IV presents the main use cases. Section V addresses open challenges, and Section VI concludes the paper.

II. 5G Network Slicing Architecture

A. End-to-End Slice Structure

A 5G network slice spans three principal domains: the Radio Access Network (RAN), the transport (midhaul/backhaul) network, and the 5G Core (5GC). Each domain contributes virtualized resources that together form an end-to-end slice, as illustrated in Fig. 1.

The 5GC adopts a Service-Based Architecture (SBA) in which network functions (NFs) such as the Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), and Network Slice Selection Function (NSSF) are exposed as microservices communicating over RESTful APIs [2]. The NSSF is particularly central to slicing: it selects the appropriate slice for each user equipment (UE) connection based on the Single Network Slice Selection Assistance Information (S-NSSAI), a standardized identifier comprising a Slice/Service Type (SST) and an optional Slice Differentiator (SD).

B. Slice Identification and Selection

When a UE requests a PDU session, it includes one or more S-NSSAIs indicating its desired slice types. The AMF forwards this information to the NSSF, which maps the request to a set of candidate Network Slice Instances (NSIs). An NSI is a running instantiation of a Network Slice Subnet—a collection of NFs and resources configured to deliver the contracted service. The AMF then routes control-plane traffic to the correct SMF within that NSI, while the SMF in turn selects the UPF that anchors the user-plane path [4].

C. RAN Slicing

At the radio access layer, slicing is realized through the functional split introduced by 3GPP in the disaggregated gNB architecture, which separates the gNB into a Central Unit (CU), a Distributed Unit (DU), and a Radio Unit (RU). Resource partitioning across slices at the DU level is handled by the RAN slice scheduler, which enforces per-slice Physical Resource Block (PRB) quotas while sharing the radio spectrum [5]. The O-RAN Alliance further extends this model through the RAN Intelligent Controller (RIC), which enables near-real-time and non-real-time control loops for dynamic slice resource adjustment.

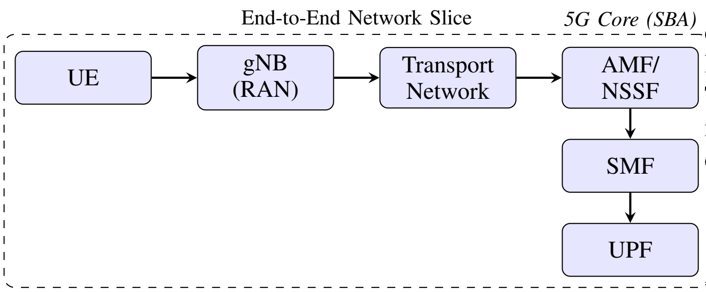

Fig. 1. Simplified end-to-end 5G network slice spanning UE, RAN, transport, and 5G Core.

III. Enabling Technologies

A. Software-Defined Networking (SDN)

SDN decouples the control plane from the data plane, allowing a logically centralized controller to program forwarding behavior across network elements via open interfaces such as OpenFlow or NETCONF/YANG. In the context of 5G slicing, SDN enables the dynamic instantiation of per-slice forwarding rules in the transport network, ensuring traffic isolation and QoS enforcement between slices without manual configuration [6]. The SDN controller also facilitates inter-domain coordination, critical for slices that span multiple operator domains.

B. Network Function Virtualization (NFV)

NFV abstracts network functions from dedicated hardware appliances and runs them as Virtual Network Functions (VNFs) or, more recently, Cloud-Native Network Functions (CNFs) deployed in containers on commodity servers [3]. The European Telecommunications Standards Institute (ETSI) NFV Management and Orchestration (MANO) framework provides the orchestration layer responsible for instantiating, scaling, and terminating VNFs in response to slice lifecycle events. CNFs leverage Kubernetes-based orchestration to enable rapid scaling and rolling updates, reducing slice provisioning time from hours to minutes.

C. Multi-Access Edge Computing (MEC)

MEC extends cloud computing capabilities to the network edge—co-located with base stations or aggregation points—thereby minimizing round-trip latency for latency-sensitive slice types [7]. In a sliced 5G network, the UPF can be deployed at the edge to enable local traffic breakout, ensuring that data never traverses the full path to a centralized data center. This is particularly important for Ultra-Reliable Low-Latency Communication (URLLC) slices used in industrial automation, where end-to-end latencies below 1 ms are required.

D. AI/ML-Driven Orchestration

Emerging standards incorporate Artificial Intelligence and Machine Learning (AI/ML) into the management plane to enable predictive slice scaling and proactive resource reallocation [8]. The 3GPP Network Data Analytics Function (NWDAF) collects telemetry from NFs and RAN nodes, trains models for load prediction, and feeds forecasts to the slice orchestrator. This closes the loop between network observation and automated adaptation, a capability essential for maintaining SLA compliance in dynamically changing traffic conditions.

IV. Use Cases

3GPP defines three canonical service categories in 5G, each naturally mapping to a distinct slice type, as summarized in Table I.

TABLE I
5G Slice Types and Representative Use Cases

Slice Type	**Service Category**	**Representative Use Cases**
eMBB	Enhanced Mobile Broadband	4K/8K video, VR/AR, fixed wireless access
URLLC	Ultra-Reliable Low Latency	Factory automation, remote surgery, autonomous vehicles
mMTC	Massive Machine-Type Comms	Smart meters, agricultural sensors, asset tracking

A. Enhanced Mobile Broadband (eMBB)

eMBB slices prioritize throughput and spectral efficiency. Typical configurations allocate large PRB quotas, enable carrier aggregation and massive MIMO beamforming, and connect to high-capacity UPFs in centralized data centers. Streaming platforms and campus networks are primary beneficiaries, as eMBB slices can deliver multi-gigabit peak rates while sharing infrastructure with other slice types [1].

B. Ultra-Reliable Low-Latency Communication (URLLC)

URLLC slices are the most technically demanding, requiring end-to-end latencies on the order of 1 ms and packet error rates below 10^{-5}. These requirements are met through short transmission time intervals (mini-slots), proactive redundancy via packet duplication (PDCP duplication), edge-deployed UPFs, and strict scheduler priority [5]. Industry 4.0 applications—including closed-loop robot control and automated guided vehicles on factory floors—are the primary drivers of URLLC slice deployment.

C. Massive Machine-Type Communications (mMTC)

mMTC slices accommodate millions of low-power devices transmitting small, infrequent payloads. These slices are optimized for connection density and energy efficiency rather than throughput. 5G NR supports mMTC through NB-IoT and eMTC enhancements, and slice configurations typically impose strict traffic policing to prevent a flood of device registrations from disrupting other slice types [4].

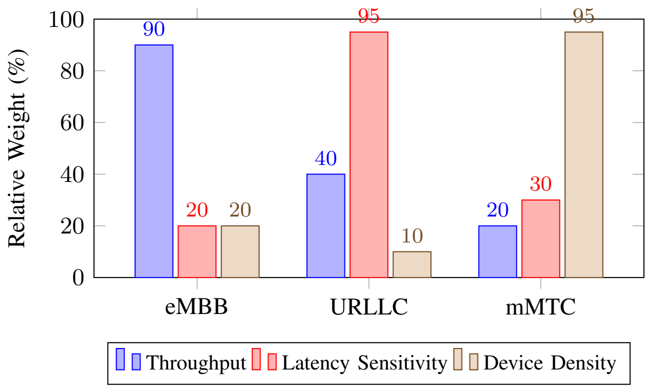

Fig. 2. Relative performance priorities across the three 5G slice types.

V. Challenges

A. Inter-Slice Isolation

Ensuring that a misbehaving or overloaded slice does not degrade neighboring slices is a non-trivial problem, particularly at the shared radio interface. Strict isolation demands per-slice scheduler enforcement, memory bandwidth partitioning in shared compute platforms (e.g., via Intel RDT), and dedicated VLAN/VxLAN segments in the transport layer. Achieving isolation without sacrificing statistical multiplexing gains remains an active area of research [6].

B. Dynamic Resource Management

Slice resource demands fluctuate with time-of-day patterns, mobility, and application behavior. Static PRB allocations waste spectrum during off-peak periods, while over-provisioning inflates operational costs. Closed-loop control driven by NWDAF predictions and RIC xApps is a promising direction, but convergence time and model accuracy under non-stationary traffic remain open challenges [8].

C. Security

Each slice represents an independent security domain, yet they share physical infrastructure. Isolation failures can expose one tenant's traffic to another. 3GPP specifies mutual authentication between UEs and slices via the Authentication Server Function (AUSF), and slice-specific security policies are enforced by the Policy Control Function (PCF). Nonetheless, side-channel attacks exploiting shared hardware resources (e.g., shared caches) pose threats that hardware-level countermeasures must address [3].

VI. Conclusion

Network slicing is the cornerstone of the 5G NR service-centric paradigm, enabling a single physical infrastructure to simultaneously support radically different application requirements through logically isolated, end-to-end virtual networks. The convergence of SDN, NFV/CNF, MEC, and AI/ML-driven orchestration provides the technical foundation for dynamic, SLA-aware slice lifecycle management. As 5G deployments mature and vertical industries deepen their reliance on sliced connectivity, addressing the remaining challenges of isolation, dynamic resource management, and security will be essential to realizing the full economic and societal potential of the technology.

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References

[1] IMT-2020 (5G) Promotion Group, "5G Concept White Paper," China Academy of Information and Communications Technology (CAICT), Feb. 2015.
[2] 3GPP, "System architecture for the 5G System (5GS)," Technical Specification TS 23.501, Rel. 17, Mar. 2022.
[3] X. Foukas, G. Patounas, A. Elmokashfi, and M. K. Marina, "Network slicing in 5G: Survey and challenges," *IEEE Communications Magazine*, vol. 55, no. 5, pp. 94–100, May 2017.
[4] 3GPP, "Procedures for the 5G System (5GS)," Technical Specification TS 23.502, Rel. 16, Jun. 2020.
[5] O-RAN Alliance, "O-RAN Architecture Description," Technical Report O-RAN.WG1.OAD-R003-v09.00, 2023.
[6] R. Mijumbi, J. Serrat, J.-L. Gorricho, N. Bouten, F. De Turck, and R. Boutaba, "Network function virtualization: State-of-the-art and research challenges," *IEEE Communications Surveys and Tutorials*, vol. 18, no. 1, pp. 236–262, First Quarter 2016.
[7] ETSI, "Multi-access Edge Computing (MEC); Framework and Reference Architecture," Group Specification GS MEC 003 V3.1.1, Jan. 2022.
[8] 3GPP, "Architecture enhancements for 5G System (5GS) to support network data analytics services," Technical Specification TS 23.288, Rel. 17, Sep. 2022.